

Ruchir Tullu

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SUMMARY

MSc physicist with 4+ years of experience developing Python software, numerical simulations, and computational tools for quantum information research. Skilled in scientific computing, algorithm development, optimization, and translating theoretical models into efficient software implementations.

EXPERIENCE

Simon Fraser University

2022 – Present

Graduate Researcher

Vancouver, BC

- Developed **Python** software implementing analytical models and numerical simulations for deterministic quantum-state preparation in permutationally invariant spin ensembles.
- Developed analytical and numerical models for optimizing entanglement-conversion protocols in continuous-variable quantum systems, resulting in a published [arXiv preprint](#).
- Designed computational tools and algorithms using **NumPy**, **SciPy**, **QuTiP**, and **Matplotlib** to perform large-scale quantum simulations, parameter sweeps, and data visualization.
- Presented research findings at the APS March Meeting (2023) and the Canadian Association of Physicists Congress (2024).

Simon Fraser University

Summer 2021

NSERC Undergraduate Student Research Assistant

Vancouver, BC

- Designed quantum and classical protocols for bosonic-to-spin state conversion.
- Implemented numerical simulations to analyze protocol efficiency across multiple parameter regimes and support theoretical development.

University of Toronto

2020 – 2021

Undergraduate Student Research Assistant

Toronto, ON

- Performed quantum-state tomography in a fibre-optics laboratory operating at telecommunications wavelengths.
- Designed and conducted experiments to quantify degradation of broadband entanglement over long-distance optical fibre and analyzed experimental measurements.

University of Calgary

2019 – 2020

Undergraduate Student Research Assistant

Calgary, AB

- Conducted finite-element simulations using **COMSOL Multiphysics** to model spin-photon quantum transducers.
- Performed density-functional-theory simulations in **Quantum ESPRESSO** to investigate candidate quantum-defect materials.

EDUCATION

Simon Fraser University

Expected Fall 2026

Master of Science, Physics (Quantum Information Theory)

Vancouver, BC

Thesis: *Deterministic Preparation of Arbitrary States in a Permutationally Invariant Spin Ensemble*

University of Toronto

Sept. 2017 – June 2021

Bachelor of Applied Science, Engineering Science (Major in Engineering Physics)

Toronto, ON

Thesis: *Experimental Study on the Degradation of Broadband Entanglement over Long Distances of Fiber*

SKILLS

Programming: Python, SQL, MATLAB, C, Bash

Scientific Computing: NumPy, SciPy, SymPy, Matplotlib

Machine Learning: PyTorch, Keras

Quantum Computing: QuTiP, Qiskit, PennyLane

Simulation Software: COMSOL Multiphysics, Quantum ESPRESSO, Mathematica

Developer Tools: Git, GitHub, Linux, Jupyter, LaTeX

Technical Expertise: Mathematical modelling, numerical simulation, optimization, scientific computing, differential equations, linear algebra, quantum information, quantum optics

AWARDS & ACTIVITIES

Awards: Mitacs Globalink Research Award (2022); Graduate Fellowship (2022); Travel & Research Awards (2022, 2024); Fluor Foundation Scholarship (2017–2021); Alexander Rutherford Scholarship (2017)

Professional Activities: Quantum Information Club, Simon Fraser University (2024–Present); American Physical Society (APS) Member (2022–Present); Canadian Association of Physicists (CAP) Member (2022–Present); Engineering Science Ambassador, University of Toronto (2018–2021); Interplanetary Space Exploration Team, University of Toronto (2018–2021)

Recognition: IBM Qiskit & Quantum Village Coding Challenge Winner (Q-SITE Conference, 2023); Poster Presentation, Quantum Machine Learning Workshop (2024)